HACKSHIFT

- **Problem Statement Title-** Sanjeevani AI – Voice-First Snakebite Emergency Response
- **Theme-** Reimagining Healthcare Through Technology
- **PS Category-** Software
- **Team Name** - Memento
- **Video Demo Link** - Click Here
- **Product Showcase Link** - Click Here

Sanjeevani AI

IDEA TITLE



Problem Statement

- **Delay Gap:** Wrong facility chosen, golden hour lost
- **Access Barrier:** No smartphone or internet in rural areas
- **Language Barrier:** Dialects unsupported on helplines
- **Stock Gap:** ASV & ICU stock unknown at PHC/CHC
- **Coordination Gap:** ER, ASHA worker & caller not linked

ACTUAL DATA STATISTICS

Deaths / year: ~58,000 (Govt data)
Bites / year: 3–4 million in India
Global share: ~50% of global deaths
Goal: Half deaths by 2030

~70% of deaths: Bihar, Jharkhand, MP, Odisha, UP +4
ASV: Polyvalent, covers “Big Four” snakes
Surveillance: Notifiable disease (Nov 2024)

PROPOSED SOLUTION

- **One-Call Response:** Dial 108 from any feature phone, no app needed
- **Voice AI Triage:** Vernacular callback scores symptom severity
- **Smart Routing:** Nearest facility with verified ASV + ICU bed
- **Parallel Dispatch:** ER doctor, ASHA worker & caller alerted together
- **First-Aid Guidance:** Voice guidance until the patient reaches care

SYSTEM ARCHITECTURE OVERVIEW

DATA & INVENTORY LAYER
ASV Stock | ICU Beds | Facilities

AI ROUTING ENGINE
Severity | Distance | Stock

LOCATION RESOLUTION
Cell Tower | Landmark | GIS

PARALLEL DISPATCH
ER Doctor | ASHA | Caller

FAST API BACKEND
Webhooks | APIs | Logic

CONTINUOUS FEEDBACK
Outcome Logs | Better Routing

USER ACCESS
Feature Phone / 108 / Missed Call

VOICE INPUT (STT)
Regional Language / Dialect

AI UNDERSTANDING
Symptoms | Landmark | Severity

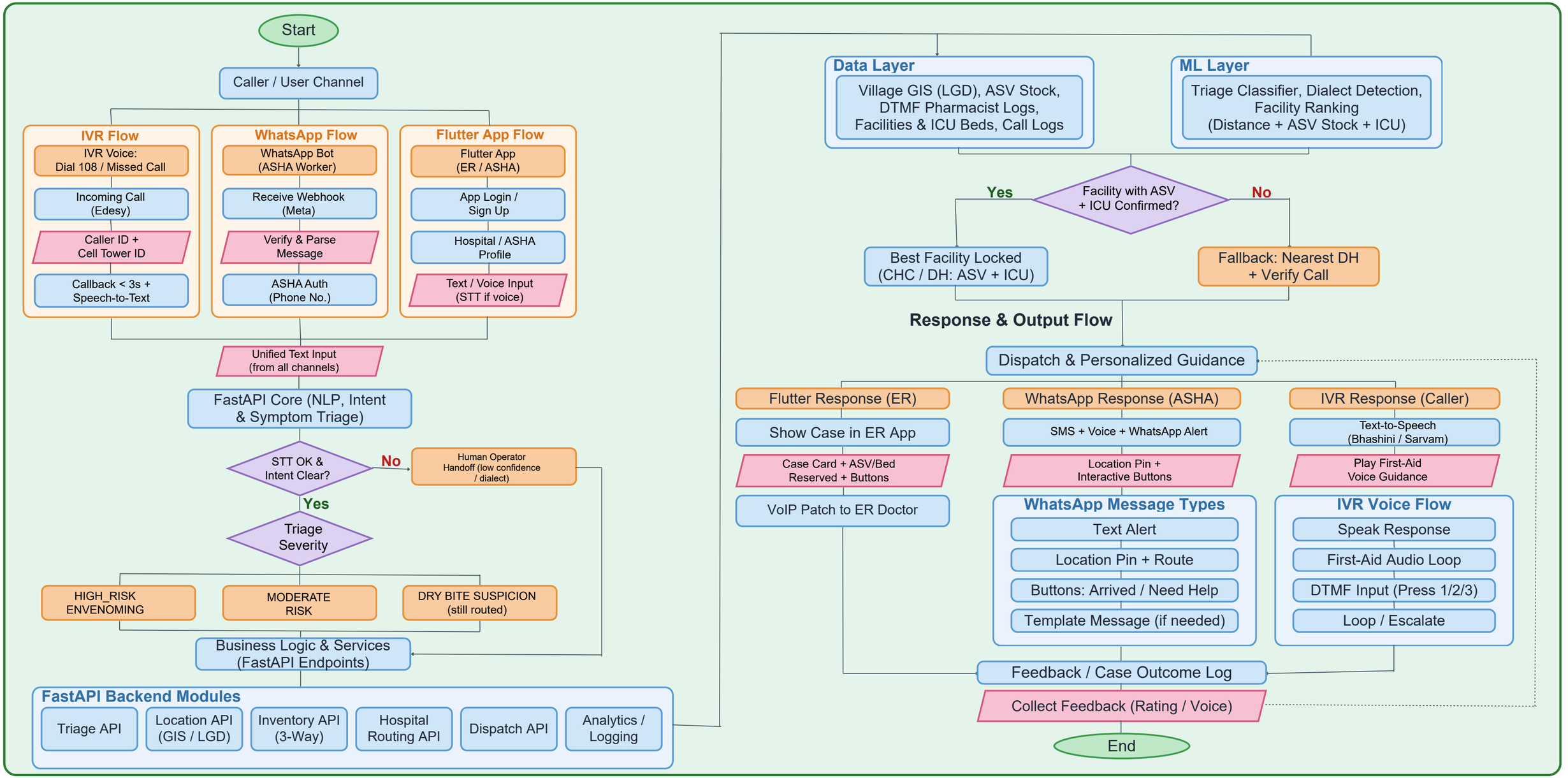
IMPROVED VERSIONS

PROJECT RESOURCES LINK

- **Figma Link** (Click here)
- **Flow Chart Design Link** (Click here)
- **Prototype Link** (Click here)

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TECHNICAL APPROACH



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FEASIBILITY & VIABILITY

FEASIBILITY

SANJEEVANI AI PROPOSED SOLUTION

- TECHNICAL FEASIBILITY
  - FastAPI
  - Voice AI APIs
  - PostGIS
- OPERATIONAL FEASIBILITY
  - Feature Phone
  - + 108 + ASHA
  - Alerts
- FINANCIAL FEASIBILITY
  - Open-source
  - Cloud-ready
  - Pay-per-use

SCALABILITY

More Districts
More Facilities
More States

IMPLEMENTABLE & SCALABLE SOLUTION

CHALLENGES

KEY CHALLENGES

- DATA GAP
  Scattered stock & bed data
- ACCESS GAP
  No smartphone or internet
- LANGUAGE GAP
  Dialects & noisy calls

→ UNIFIED & INCLUSIVE SOLUTION

VIABILITY

VIABILITY

- USER ADOPTION
  Zero install
  Dial 108
- SUSTAINABILITY
  Low-cost
  Open-source
- SCALABILITY
  More districts
  More states

→ LONG TERM IMPACT
Fewer Snakebite Deaths

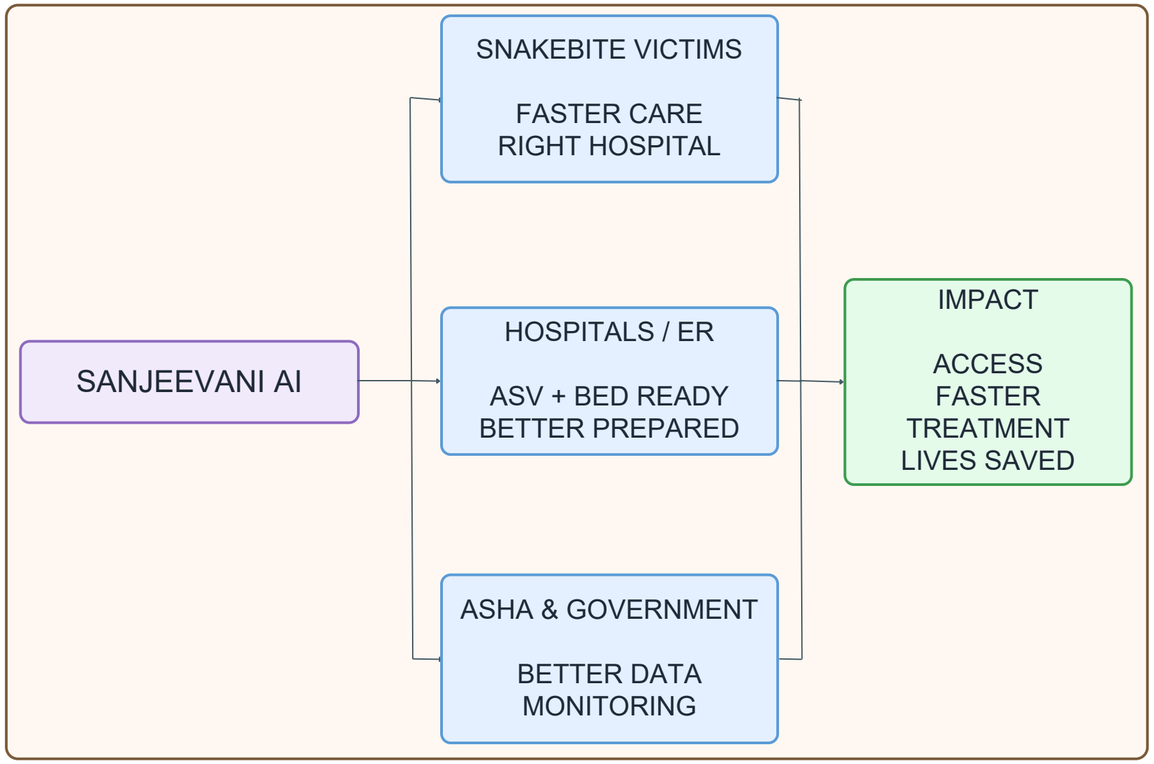
OUR STRATEGIC APPROACH

- DATA FOUNDATION
  - Consolidate facility, ASV stock, ICU bed & Panchayat GIS data into one verified inventory
- MULTILINGUAL AI
  - Understand regional languages & dialects with Bhashini / Sarvam speech models
- SYMPTOM + LANDMARK EXTRACTION
  - Identify symptoms, severity cues and village / landmark from the caller's speech
- INTELLIGENT ROUTING
  - Match caller severity & location with the nearest facility holding verified ASV + ICU
- PARALLEL DISPATCH
  - One backend alerts ER Doctor, ASHA Worker & Caller at the same time
- VALIDATION & CONTINUOUS IMPROVEMENT
  - Simulated calls → Dialect tests → Clinician review → Pilot district

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IMPACTS & BENEFITS

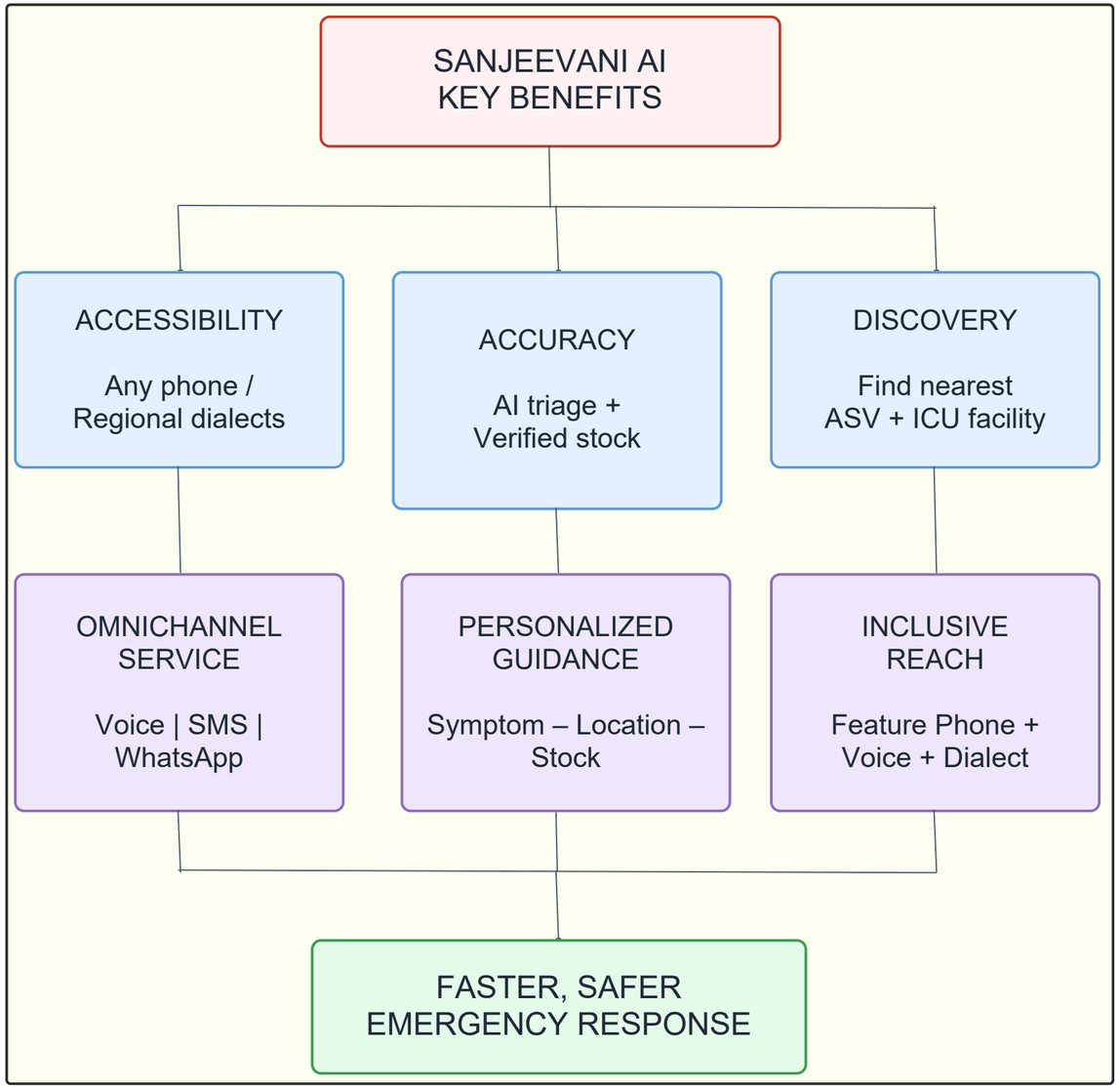
TARGET AUDIENCE IMPACTS



➔ Turns a panicked 108 call into fast, location-aware emergency routing.
➔ Connects every victim with the nearest facility that actually has ASV and an ICU bed.

➔ Strengthens inclusive emergency access for feature-phone and rural users.
➔ Supports the NAPSE goal of halving snakebite deaths by 2030.

BENEFITS



RESEARCH & REFERENCES

RESEARCH REFERENCES AND DATABASE

- National Hospital Directory with Geo Code and Additional Parameters (Reference Link)
- Official snakebite guidance — for the IVR conversation (Reference Link)
- Standard Treatment Guidelines for Management of Snake Bite (Reference Link)
- National Programme for Prevention and Control of Snakebite Envenoming — NCDC (Reference Link)
- C-DAC — e-Aushadhi Healthcare Informatics (Reference Link)